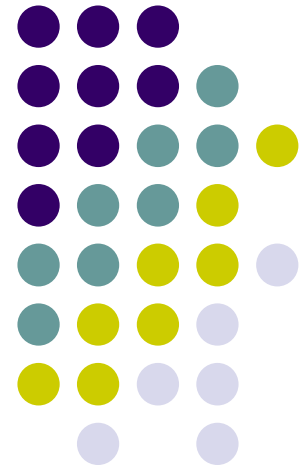
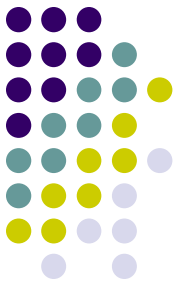


Behaviour modelling 2

Dynamic Modelling for
Human-centered Systems

Chapter 10 – 11 – 12

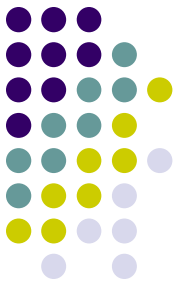




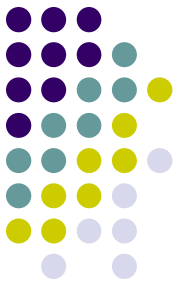
What should you learn?

- Different types of behaviours, which require types of cognitive complexity
 - Environmental complexity thesis
- Logical paradigm forms a **flexible** way to describe those
- How to make such models
 - conceptual model
 - ontology (predicates, sorts), with **standard** terms
 - generic FOR EVERY TIMEPOINT rules
 - semi formal + formal notation

Overview of modelling elements

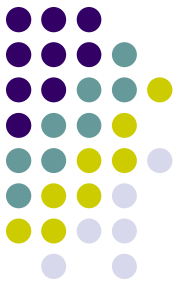


- **predicates**: the names used for the concepts and their argument types
 - **observed**: AGENT x OBJECT
 - "observed": ["AGENT", "OBSERVABLE"],
- **sorts**: the atomic values for arguments types
 - **AGENT**: arnie, bernie, charlie
 - "AGENT": ["arnie", "bernie"],
 - **OBSERVABLE**: screen, food_present
 - "OBSERVABLE": ["screen", "food_present"],



Types of behaviour

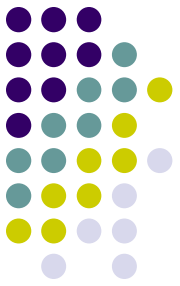
- Previous lecture
 - Reactive behaviour
 - Stimulus-response behaviour
 - Delayed-response behaviour
- This lecture
 - Motivation-based behaviour (Chapter 10)
 - BDI-model
 - Adaptive behaviour (Chapter 11)
 - Extended mind behaviour (Chapter 12)



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Chapter 10

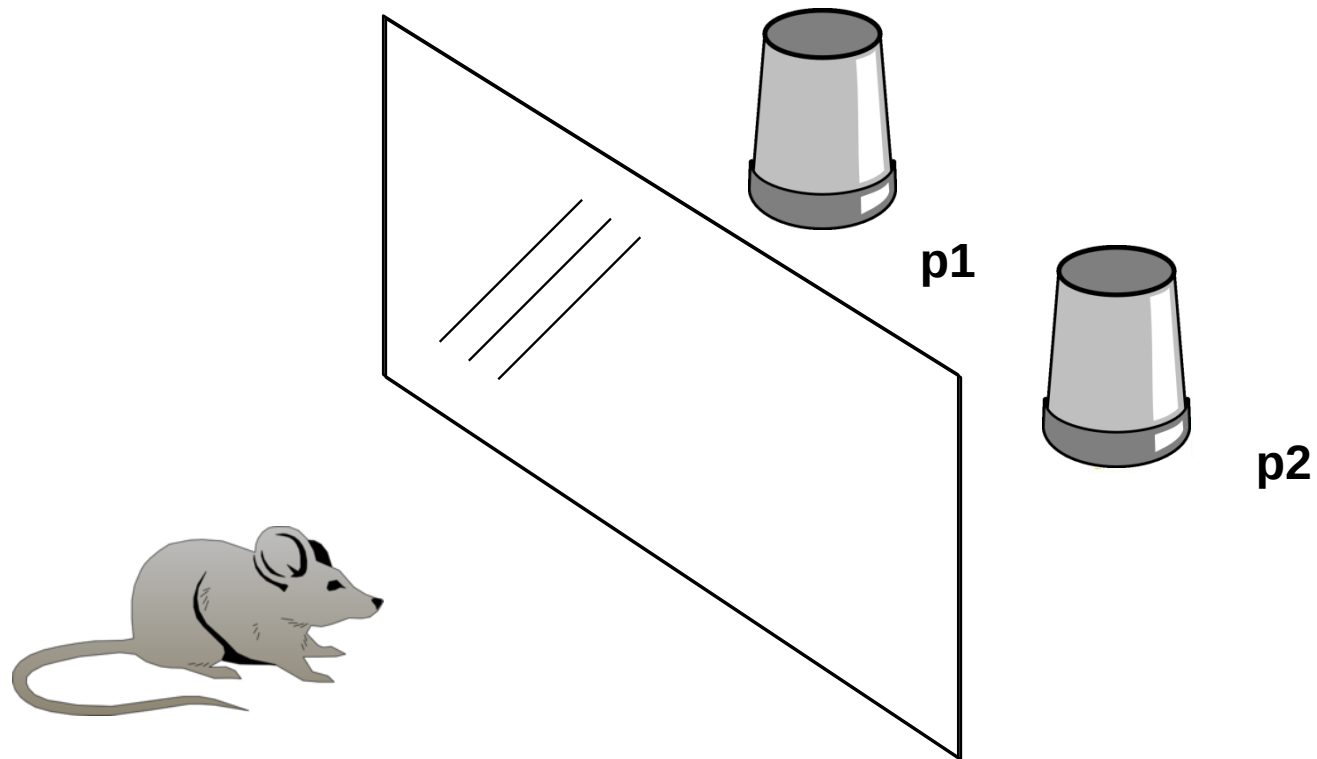
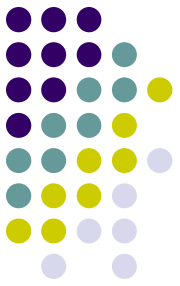
MOTIVATION-BASED BEHAVIOUR



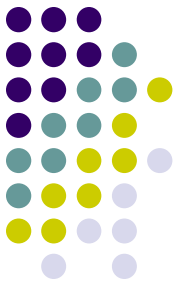
Motivation-Based Behaviour

- Patterns of behaviour in which the animal
 - takes the **initiative**
(without observable direct reason)
- Viewed from external and internal viewpoint

Experimental Setting

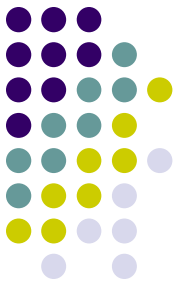


Motivation-Based Behaviour: Example Traces



	<i>time 0</i>	<i>time 1</i>	<i>time 2</i>	<i>time 3</i>	<i>time 4</i>	<i>time 5</i>
<i>trace 1</i>	food screen	no food no screen	food screen	food screen	food no screen	food no screen goto p2
<i>trace 2</i>	no food no screen	food no screen	no food screen	no food no screen	no food no screen	no food no screen
<i>trace 3</i>	no food no screen	no food no screen	food screen	food no screen	food no screen	food no screen
<i>trace 4</i>	no food no screen	food no screen	no food screen	food screen	food no screen	food no screen goto p2

External explanation of Motivation-based behaviour



- Why does the animal go to p2 ?
 - The animal starts wanting food if it cannot get it; it then reacts immediately if food is present and screen is removed

The animal goes to p2, because

it just observed the absence of the screen, and at an earlier point in time

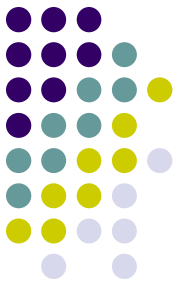
it observed food at p2,

whereas at a still earlier point in time

it observed the absence of food at p2, and at yet another still earlier point in time

it observed the presence of food at p2.

Motivation-based behaviour: external description (1)



EMB1

at any time point $t1$

if the animal observes the absence of the screen,

and there exists time points $s1$, $s2$, and $s3$

with $s1 < s2 < s3 \leq t1$ such that

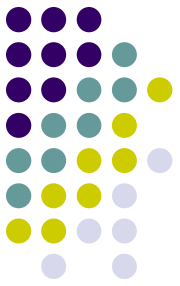
at $s3$ the animal observed the presence of food at $p2$,

and at $s2$ the animal observed the absence of food at $p2$,

and at $s1$ the animal observed the presence of food at $p2$,

then after time point $t1$ the animal will go to $p2$

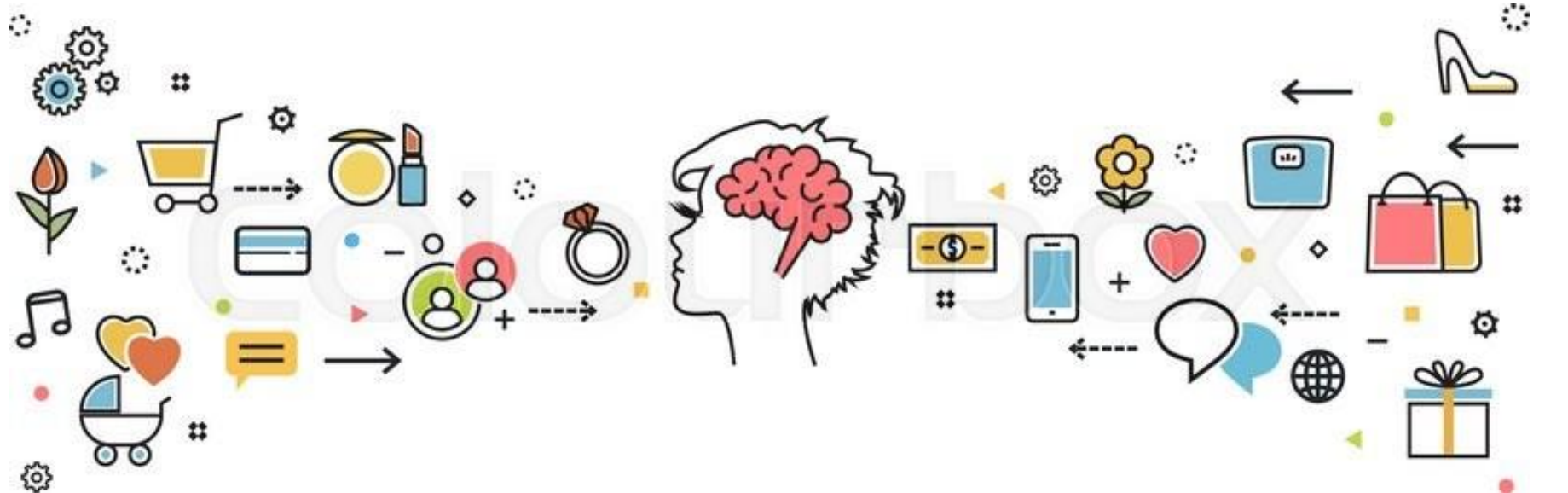
External Explanation of Motivation-Based Behaviour (2)



- More a **historical** type of explanation than based on a direct cause
- Doubtful whether this is a **satisfactory** explanation
- For an explanation based on direct causes internal state properties seem to be required

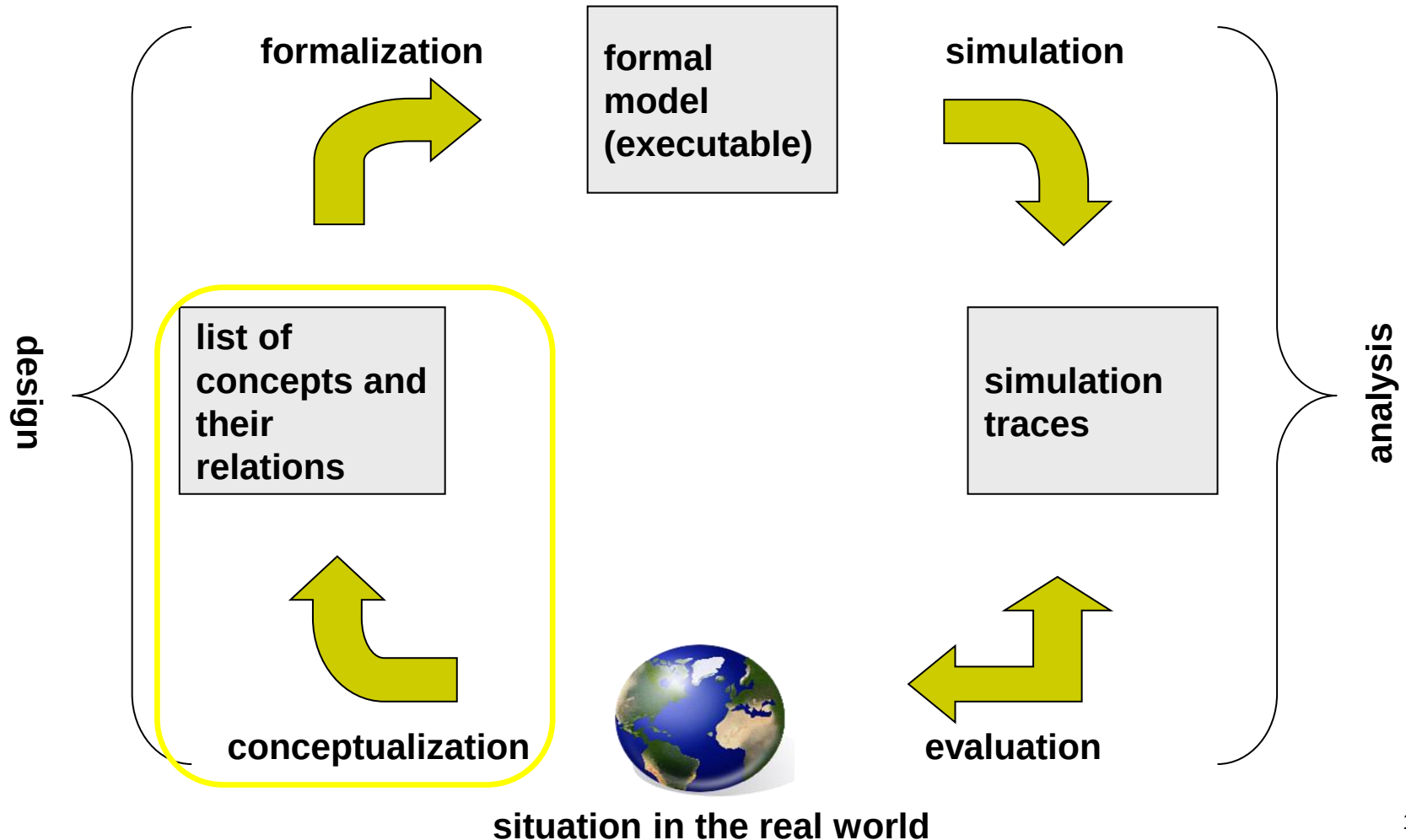
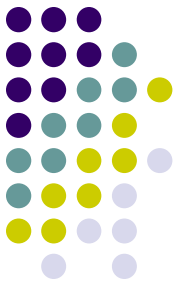
because of reference to past input

Internal Explanation

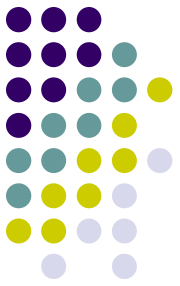


- **Desires** steer behaviour
 - mediated by beliefs
 - lead to intentions

Modelling Cycle: Conceptualization



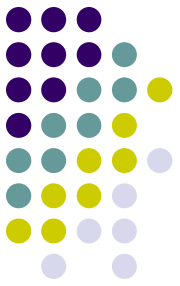
Specification of input and output



Input: **observation**
(of presence of food and screen)

Output: **action**
(of going to p2)

Specification of internal cognitive states

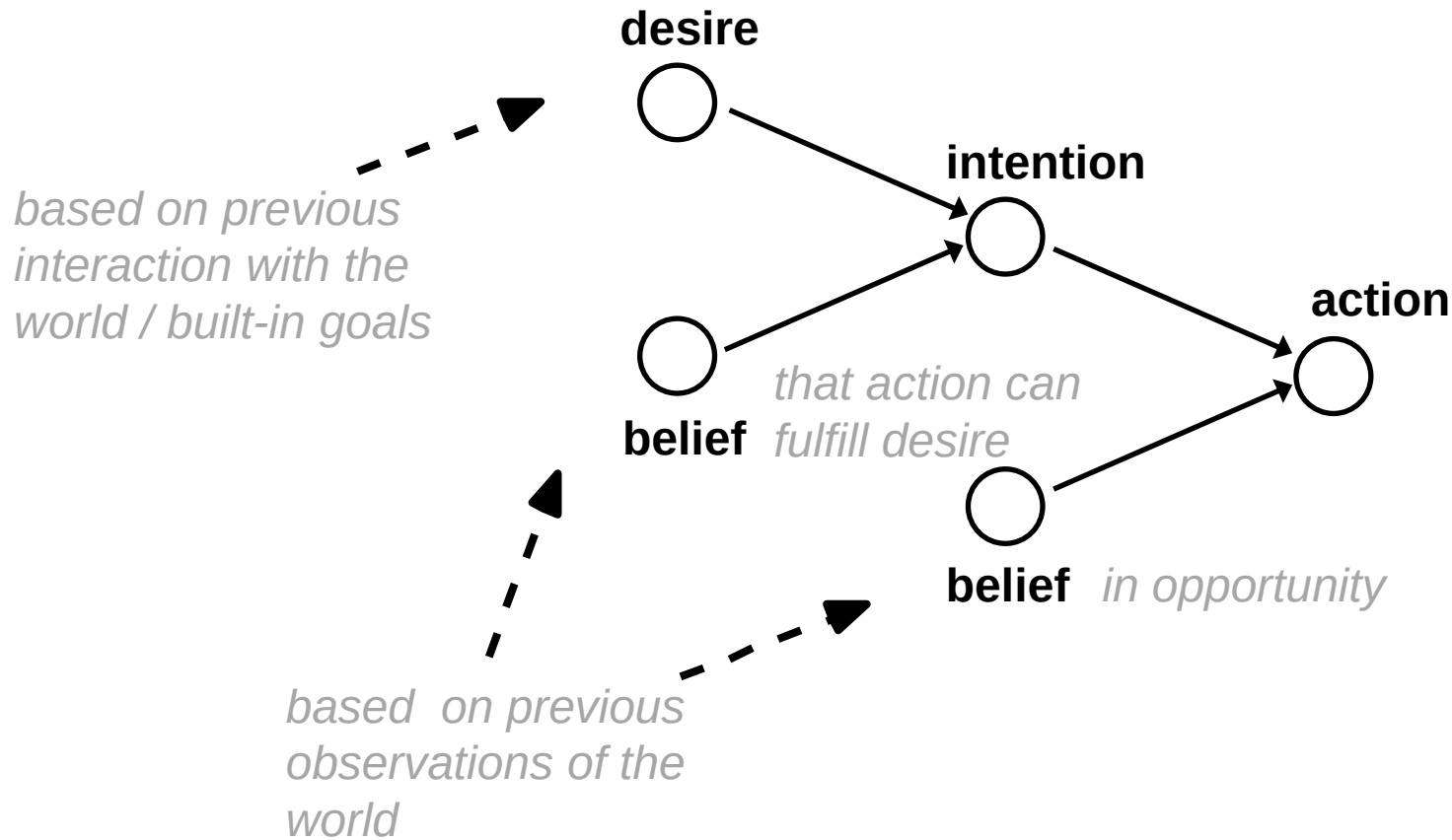
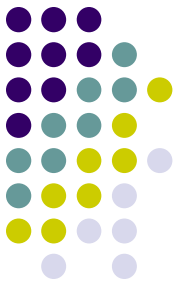


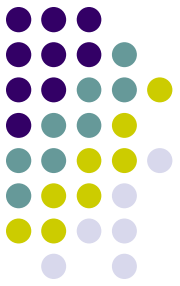
Internal: **belief**
(on presence of food and screen)

desire
(to have food)

intention
(to go to p2)

BDI Model

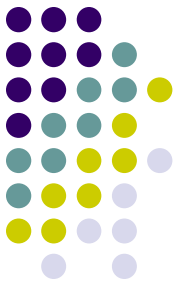




Assignment

Try to develop a (graphical) BDI model for the
'motivation-based mouse'

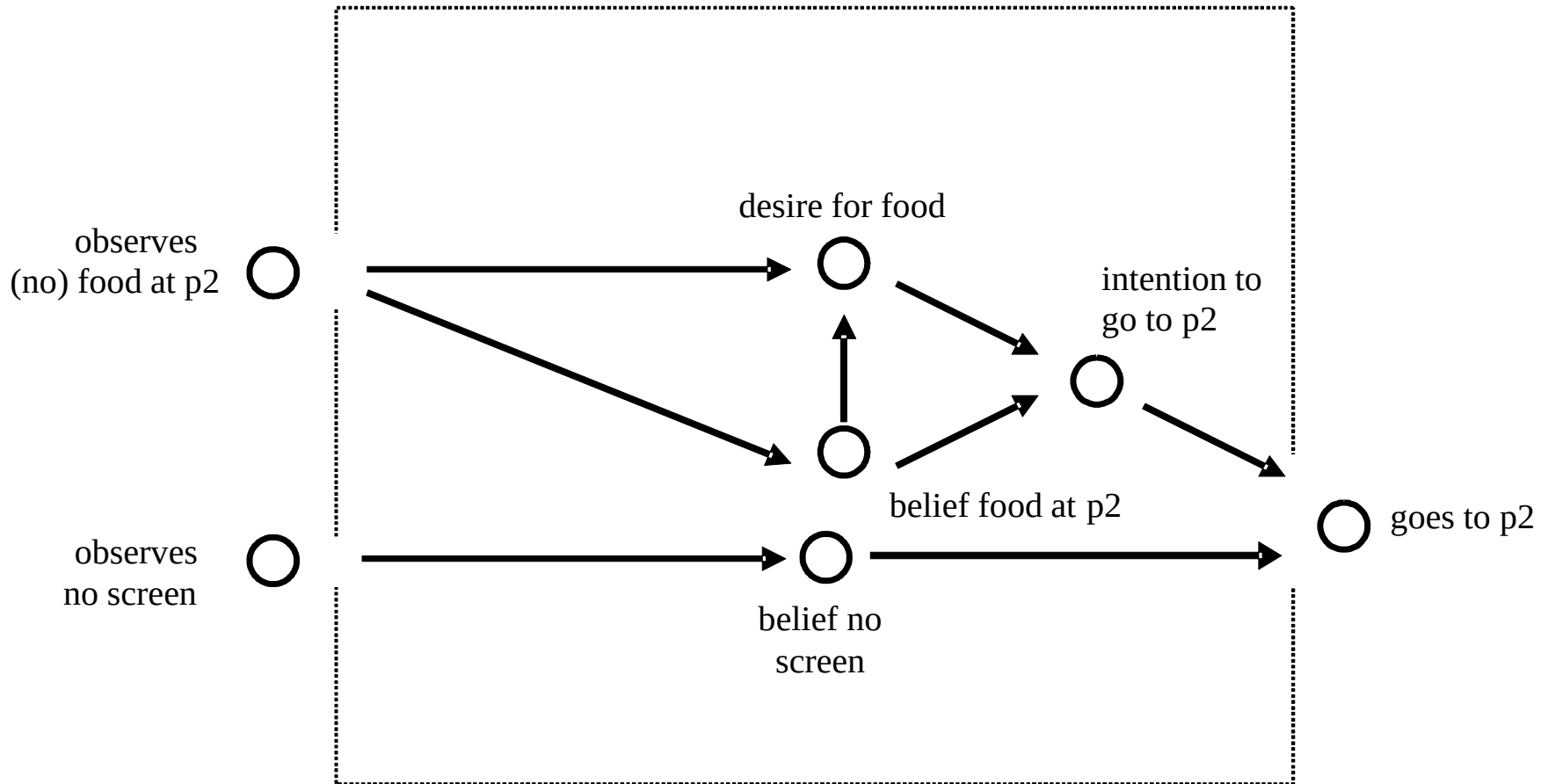
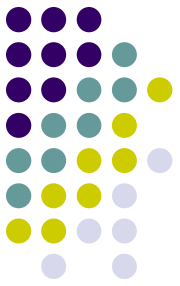
Use the concepts given on the next slides

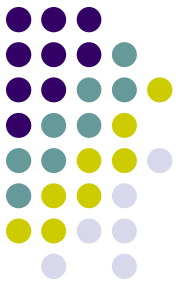


Concepts

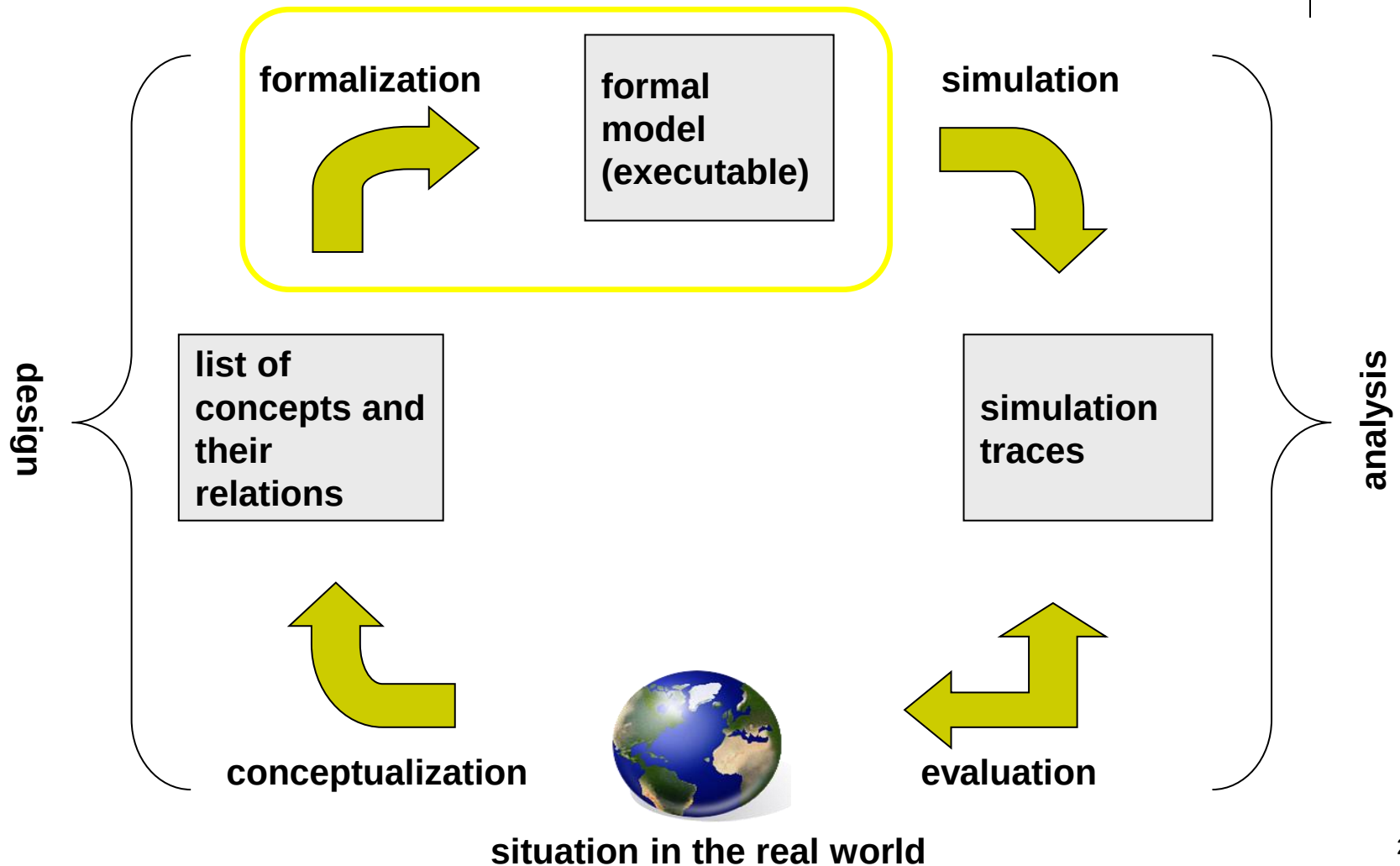
- observes_food_at_p2
- observes_no_food_at_p2
- observes_no_screen
- belief_no_screen
- belief_food_at_p2
- desire_for_food
- intention_to_go_to_p2
- goes_to_p2

Relationships between concepts



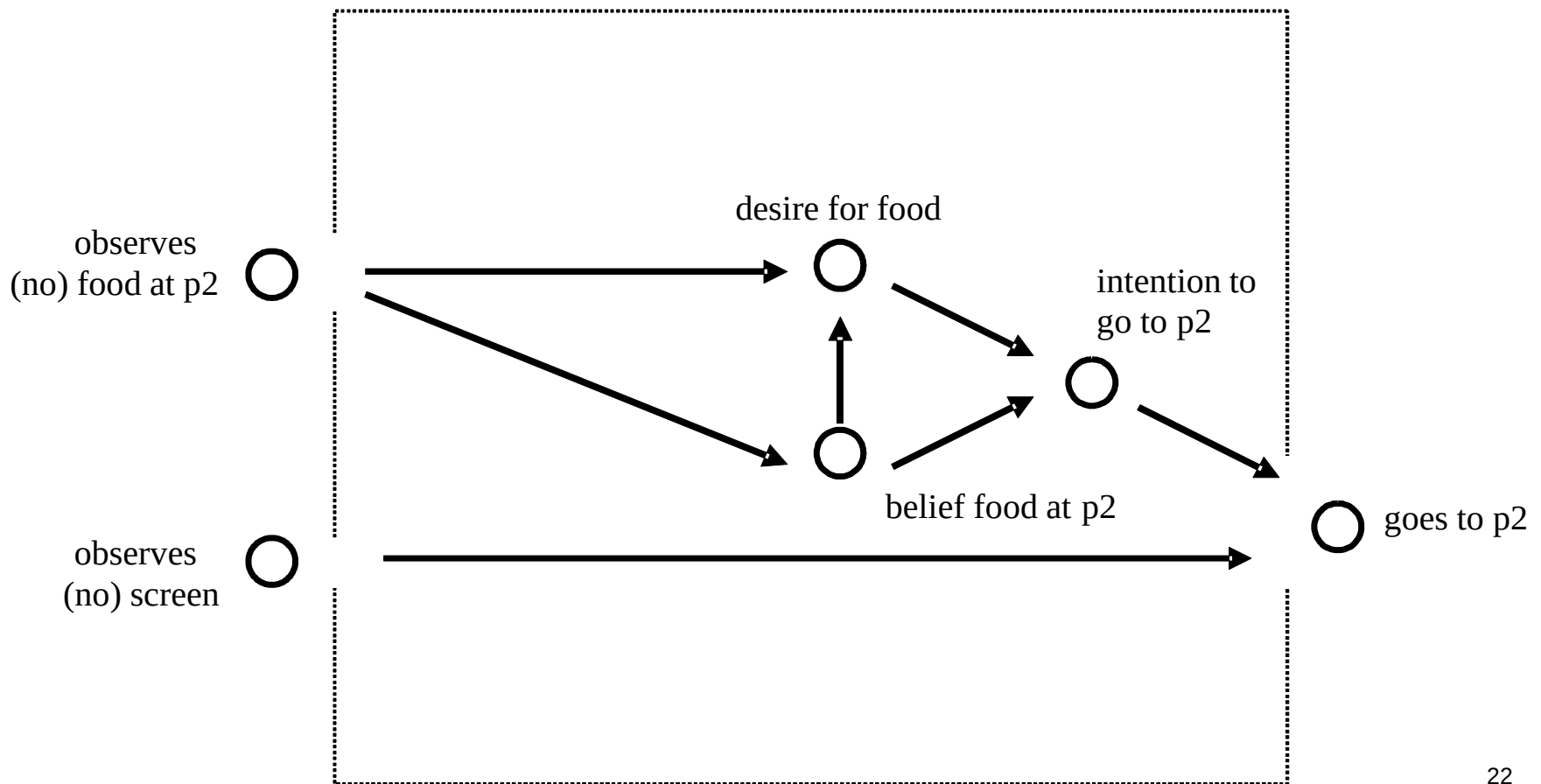
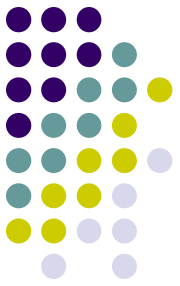


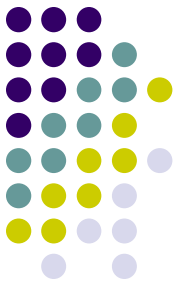
Modelling cycle: Formalization



Relationships between concepts

without “belief screen”



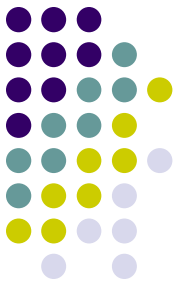


The Detailed Model

Formalisation of the concepts:

External behavioural level	
observation (of presence of food or screen)	
action (go to p2)	
Internal cognitive level	
belief (of presence of food)	
desire (to have food)	
intention (to go to p2)	

Motivation-Based Behaviour: Internal Dynamics (1)



IDR1 (Belief Generation)

for all time points

if the animal observes that food is present at $p2$,
then the belief that food is present at $p2$ will hold

$\text{observed}(\text{at_position}(\text{food}, p2)) \longrightarrow \text{belief}(\text{at_position}(\text{food}, p2))$

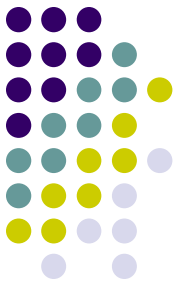
IDR4 (Belief Persistence)

for all time points $t1$ and $t2$ with $t1 < t2$

if the belief that food is present at $p2$ holds at $t1$,
and between $t1$ and $t2$ the agent does not observe
that food is not present at $p2$,
then the belief that food is present at $p2$ holds at $t2$

$\text{belief}(\text{at_position}(\text{food}, p2)) \wedge \neg \text{observed}(\text{not}(\text{at_position}(\text{food}, p2)))$
 $\longrightarrow$
 $\text{belief}(\text{at_position}(\text{food}, p2))$

Motivation-Based Behaviour: Internal Dynamics (2)



IMB1 (Desire Generation)

at any point in time

if the animal observes the absence of food at $p2$,

and the belief that food is at $p2$ holds

then the desire for food will hold

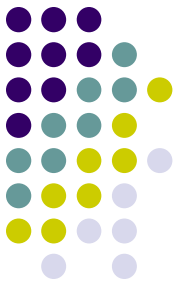
$\text{observed}(\text{not}(\text{at_position}(\text{food}, p2))) \wedge \text{belief}(\text{at_position}(\text{food}, p2))$



$\text{desire}(\text{have_food})$

Motivation-Based Behaviour:

Internal Dynamics (3)



IMB2 (Intention Generation)

at any point in time

if the desire for food holds,
and the belief food is present at $p2$ holds
then the intention to go to $p2$ will hold

$\text{desire}(\text{have_food}) \wedge \text{belief}(\text{at_position}(\text{food}, p2) \longrightarrow \text{intention}(\text{goto}(p2))$

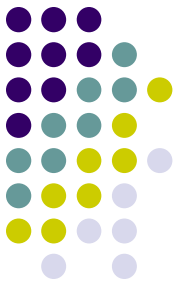
IMB3 (Action Generation)

at any point in time

if the intention to go to $p2$ holds,
and the animal observes the absence of the screen
then the animal will go to $p2$

$\text{intention}(\text{goto}(p2)) \wedge \text{observed}(\text{no_screen}) \longrightarrow \text{performed}(\text{goto}(p2))$

Motivation-Based Behaviour: Internal Dynamics (4)



IMB4 (Desire Persistence)

for all time points

if internal cognitive state property desire for food holds,
then for every later time point
 internal cognitive state property desire for food holds

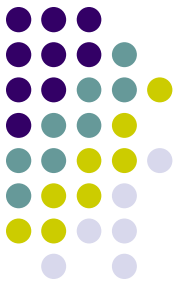
desire(have_food) $\longrightarrow$ desire(have_food)

IMB5 (Intention Persistence)

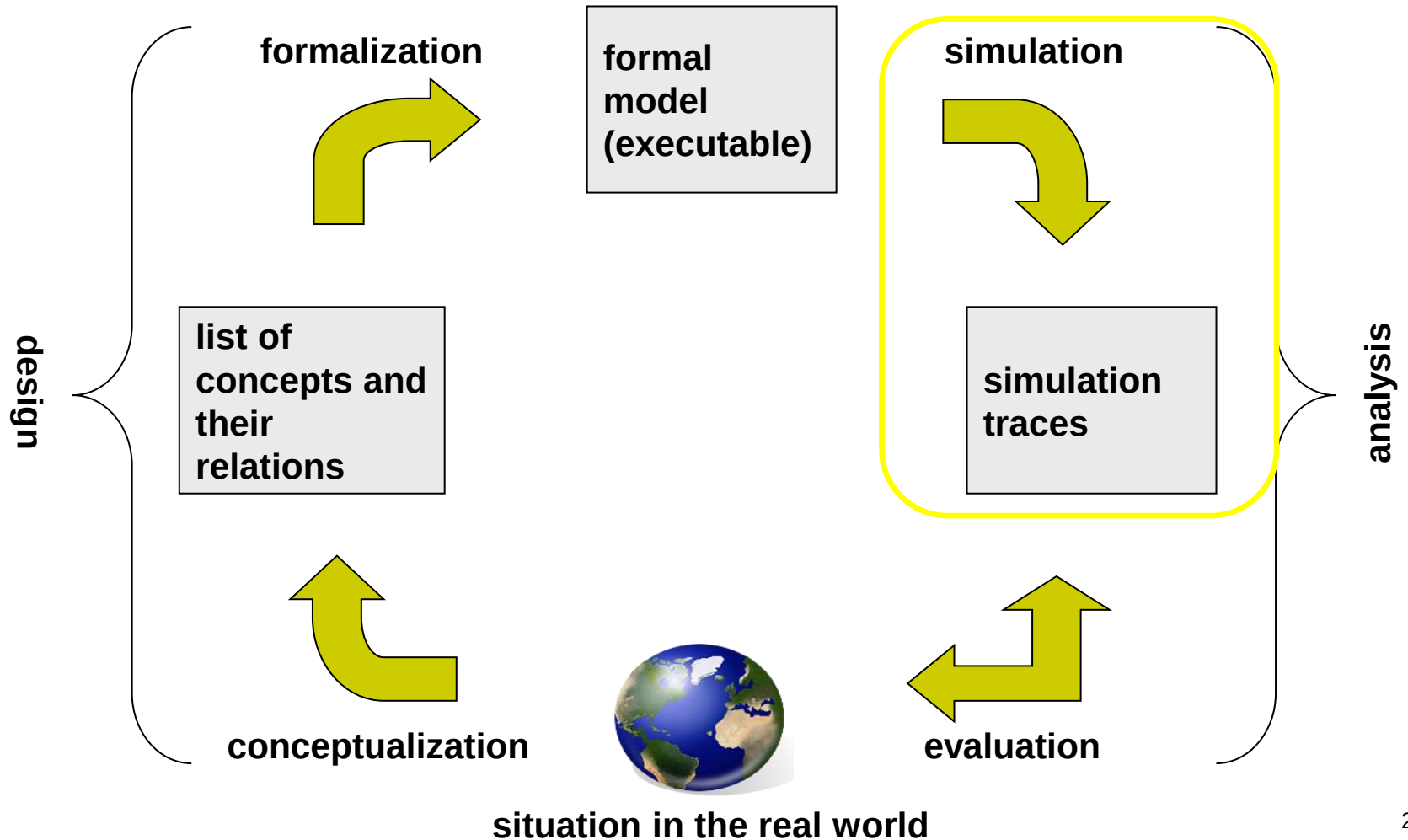
for all time points

if internal cognitive state property intention to go to p2 holds,
then for every later time point
 internal cognitive state property intention to go to p2 holds

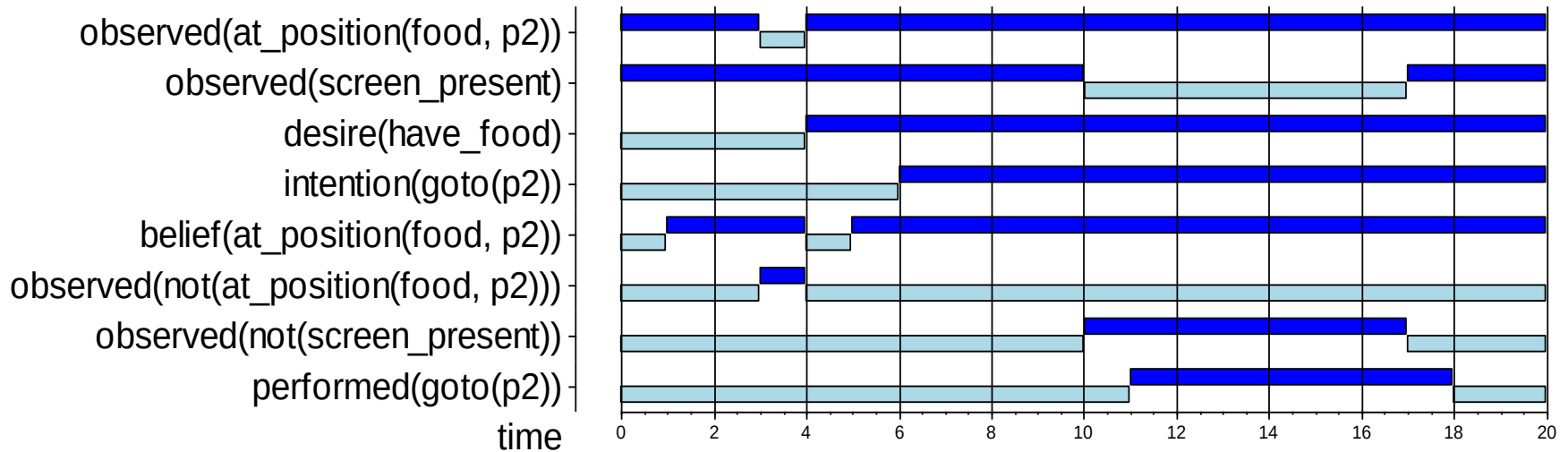
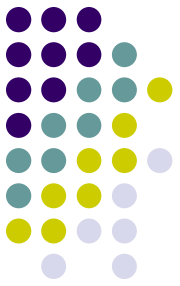
intention(goto(p2)) $\longrightarrow$ intention(goto(p2))



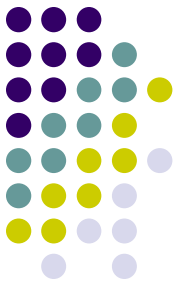
Modelling cycle: Simulation



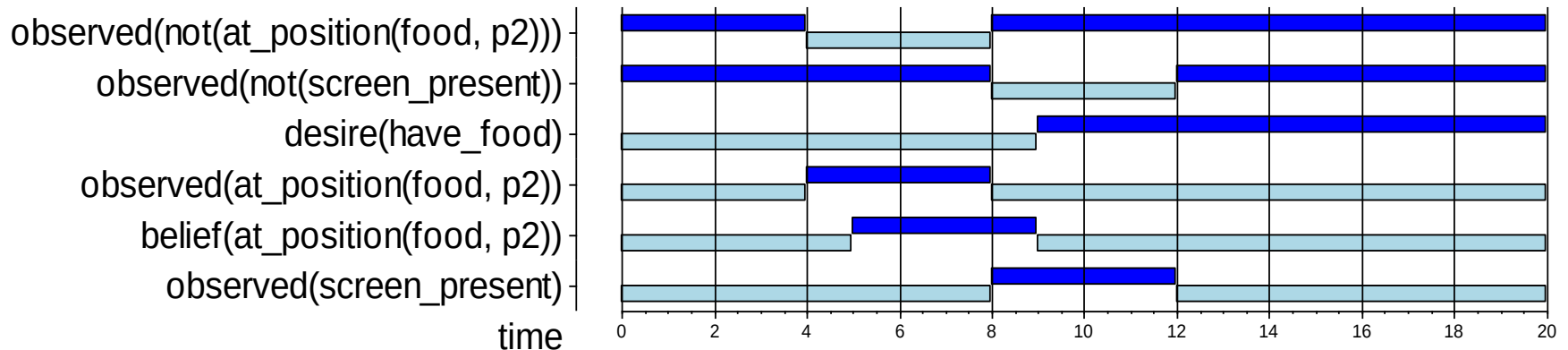
Example Simulation Traces (1)



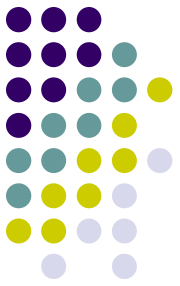
Example Simulation Traces (2)



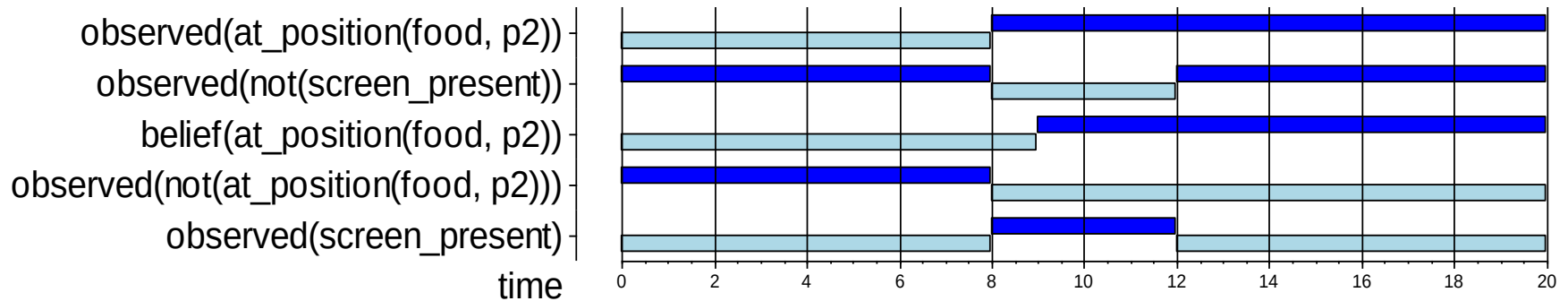
- no **intention** generated



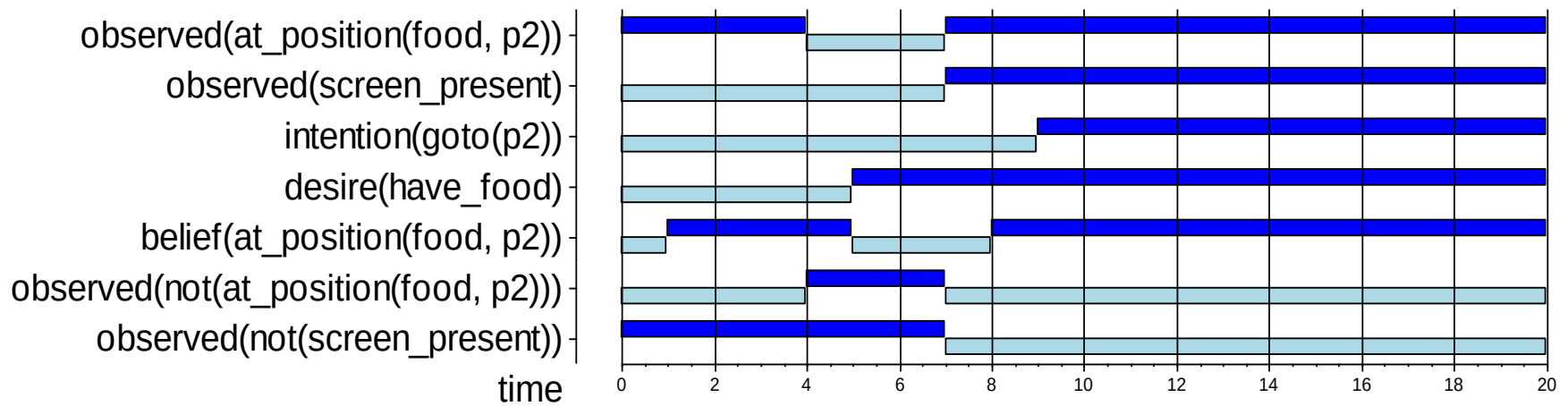
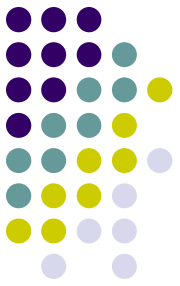
Example Simulation Traces (3)



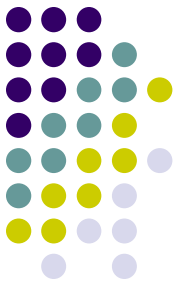
- no **desire** generated



Example Simulation Traces (4)

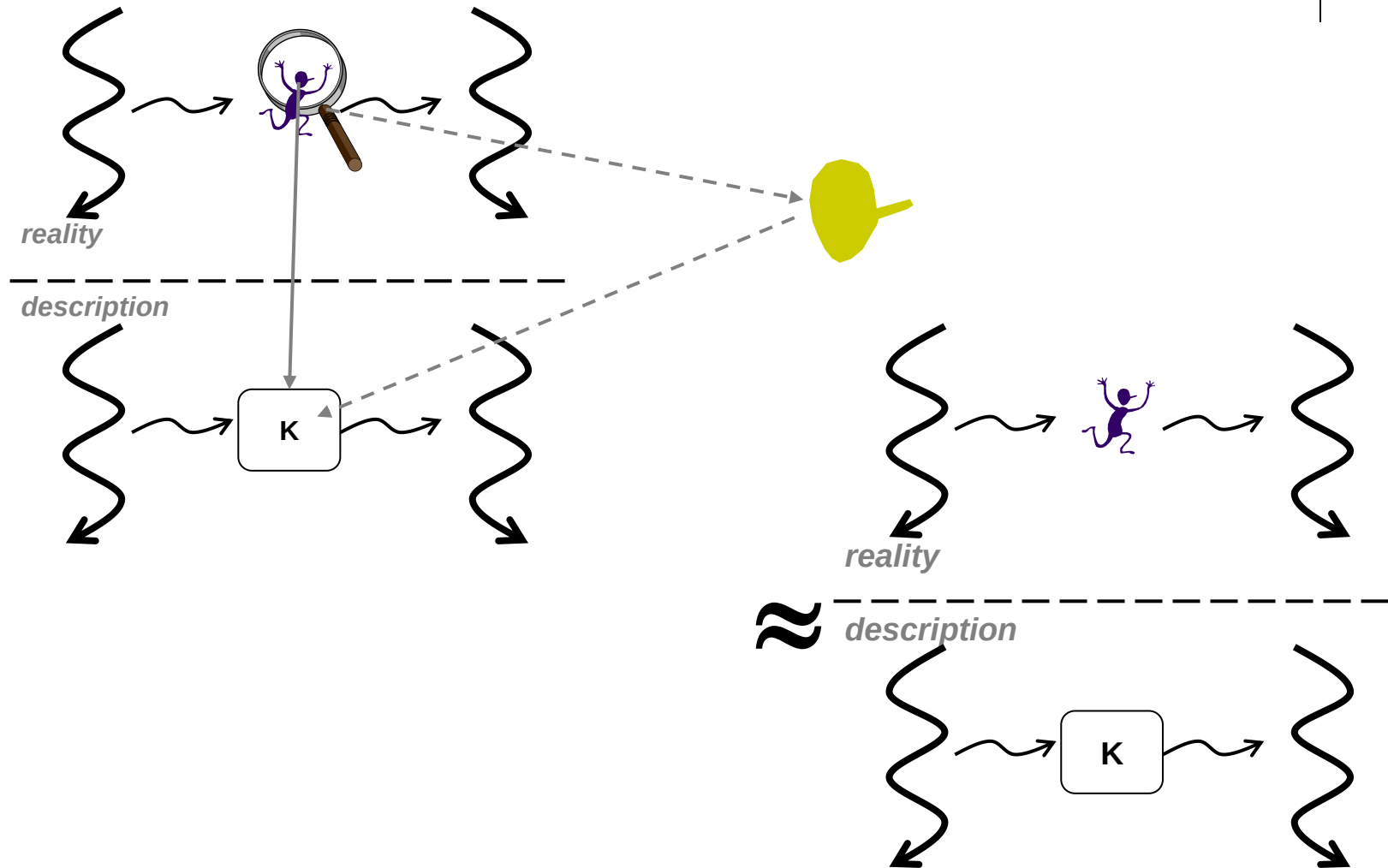
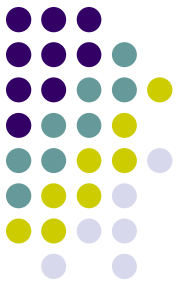


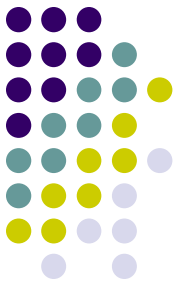
Motivation-based behaviour: Summary



- Patterns of behaviour in which the animal takes the **initiative**
- External perspective: may require nontrivial references to past input and/or output
- Internal perspective: using **internal state properties** such as **beliefs**, **desires**, **intentions**

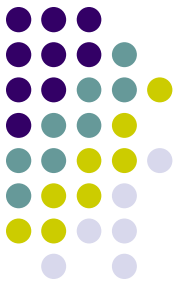
What are the internal states?





What are the internal states?

- Other terms for those two perspectives:
 - Realism:
 - The internal cognitive state properties are (known to be) related to certain ‘**real**’ physical (e.g., neurological) state properties.
 - Instrumentalism
 - The internal cognitive state properties are not (known to be) related to ‘real’ physical state properties. They are only **instruments**, e.g., to do predictions on behaviour.



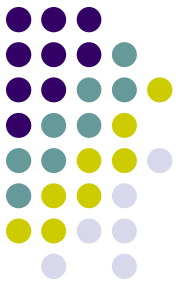
Intentional Stance

A specific instance of an **instrumentalist** view is Dennett's **intentional stance**:

- attribution of intentional state properties to others
- intentional strategy to explain and predict behaviour

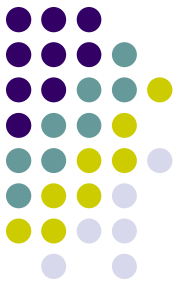
(Dennett, 1987)





Intentional Stance

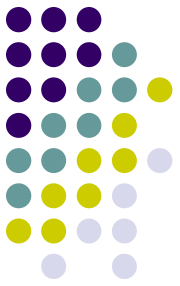
- Treat the other as a rational **individual**
- Figure out what **beliefs** it has,
given its place in the world and its purpose
- Figure out what **desires** it ought to have,
given its place in the world and its purpose
- **Predict** that the individual will behave
according to its desires and beliefs



Physical Stance

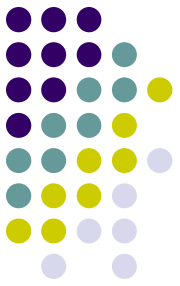
- This can be opposed to the **physical stance** (**realist** view):
 - determine the physical constitution of a system
 - use knowledge of **laws of physics** to predict the behaviour depending on the inputs
 - often useful for engineers

Advantages of Int. Stance (1)

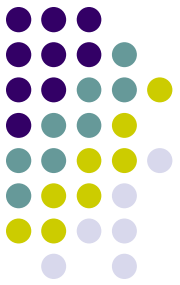


‘Predicting that someone will duck if you throw a brick at him is easy from the folk-psychological stance; it is and will always be intractable if you have to trace the photons from brick to eyeball, the neurotransmitters from optic nerve to motor nerve, and so forth.’
(Dennett, 1991), p. 42

Advantages of Int. Stance (2)

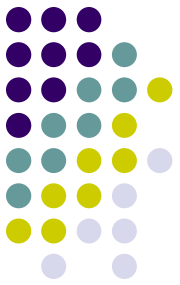


- An intentional stance description **abstracts** from specific aspects of physical realisation; therefore
 - the description is more **concise** and **transparent**
 - it is more **generic**: it applies to more cases, in different physical realisations



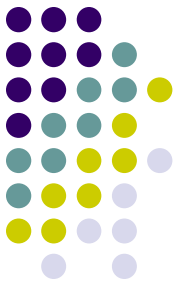
Conclusion

- Humans often try to **understand** and **influence** others by reasoning about other human's **beliefs**, **desires** and **intentions**
 - this is crucial for functioning in human society
- Intentional stance is philosophical basis for intelligent systems
 - if we want to build systems that “understand” people, it is useful to model their behaviour from an intentional stance



Chapter 11

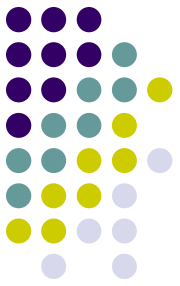
ADAPTIVE BEHAVIOUR



Adaptive Behaviour

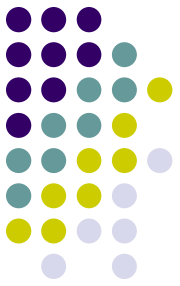
- Patterns of behaviour in which the individual after some point in time shows different behaviour than before, due to, for example,
 - learning
 - trauma
 - (forgetting)
- This new behaviour is persistent over a longer time period

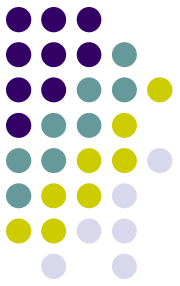
Environmental Complexity Thesis



- **stimulus-response** behaviour
 - works only with visible food
- **delayed response** behaviour
 - food becoming invisible is taken into account
- **motivation-based** behaviour
 - food becoming absent is taken into account
 - never visible food is taken into account
- **adaptive behaviour**
 - accommodate behaviour to environment

Example

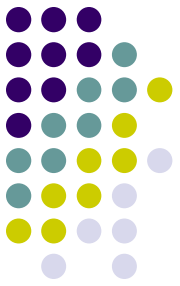




Wasp Example: Before

- Dynamic relationship satisfied by traces before the adaptation:

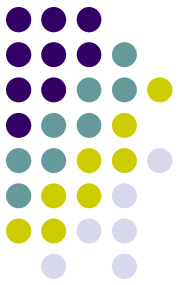
for all time points until some time point t
if the individual observes that a wasp **is close** to the individual,
then the individual will **not panic**



Wasp Example: After

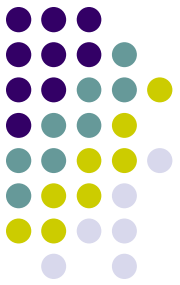
- Dynamic relationship satisfied by traces after the adaptation (adapted behaviour):

for all time points after some time point t
if the individual observes that a wasp **is**
close to the individual,
then the individual will **panic**



Wasp Example: The Process

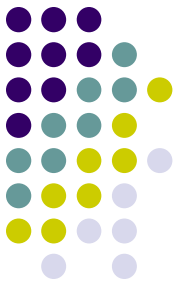
- What happened to you that makes you behave in this way?
- Why are you the way you are ?
 - ‘behaviour after’ compared to ‘behaviour before’ shows the result of adaptation but does not show the adaptation process.
- Which dynamic relationships characterise the adaptation process itself ?



Example Traces

	<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
<i>trace 1</i>	wasp close	wasp close	wasp stings	wasp close ouch!	wasp close panic	no wasp panic	no wasp	wasp close	wasp close panic
<i>trace 2</i>	wasp close	wasp close	no wasp	no wasp	wasp close	wasp close	no wasp	wasp close	wasp close
<i>trace 3</i>	wasp close	wasp stings	no wasp ouch!	no wasp	wasp close	wasp close panic	no wasp panic	wasp close	wasp close panic
<i>trace 4</i>	wasp close	wasp close	no wasp	wasp close	wasp stings	wasp close ouch!	no wasp panic	wasp close	wasp close panic

External Explanation of Adaptive Behaviour



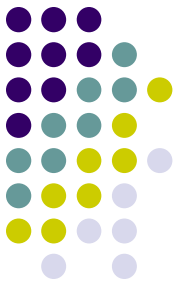
Suppose a trace is given:

	<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
<i>trace 4</i>	wasp close	wasp close	no wasp	wasp close	wasp stings	wasp close ouch!	no wasp panic	wasp close	wasp close panic

Why does the individual panic at time point 8 ?

The individual panics, because ...

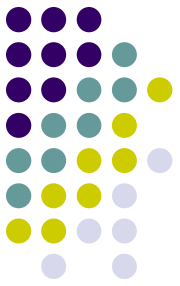
Specification of Input and Output



Input: **observation** of stimuli
(of a wasp being close)
experience
(of being stung)

Output: **action**
(of: panicking, screaming “ouch”,
doing nothing)

Specification of Internal Cognitive States

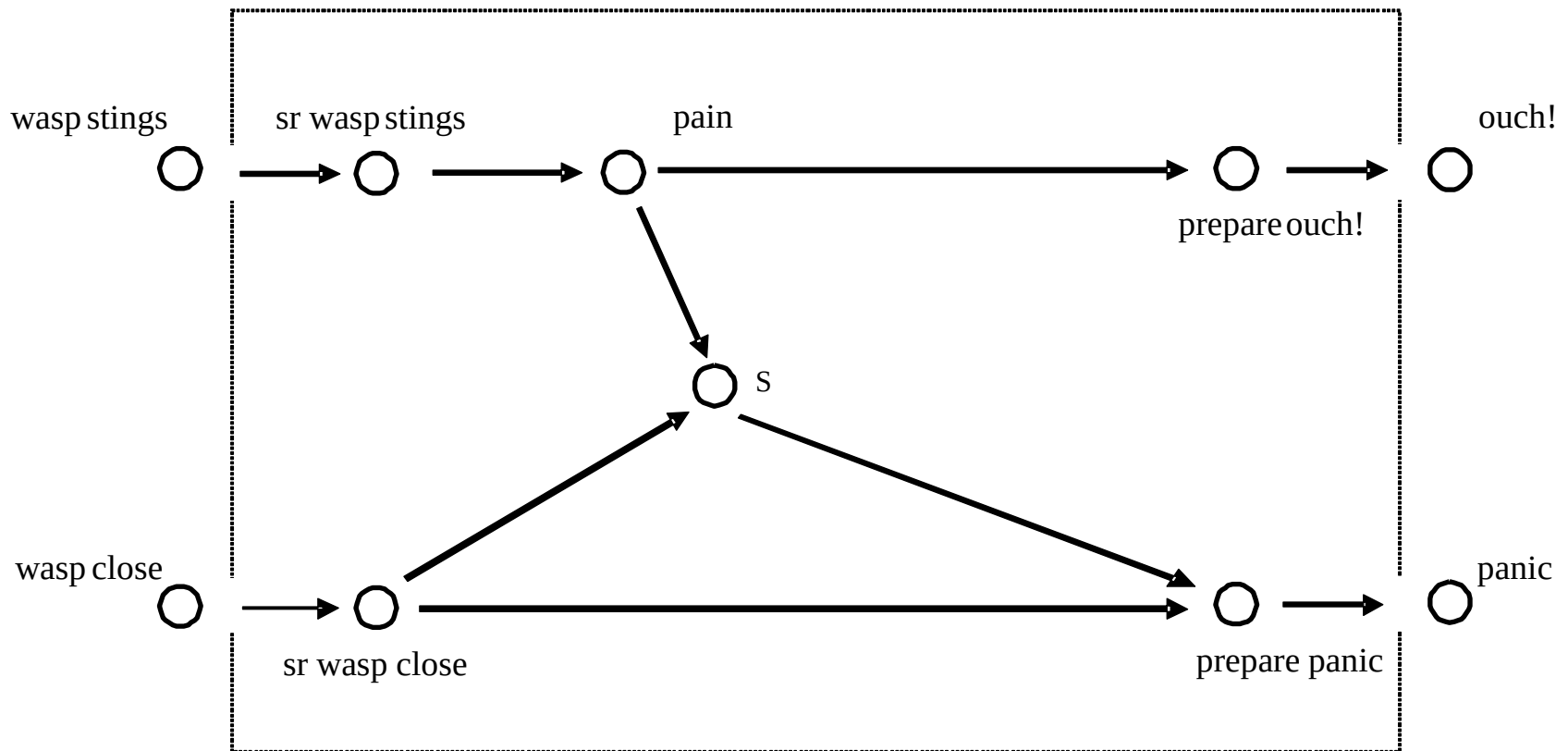
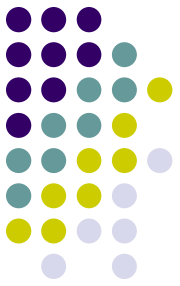


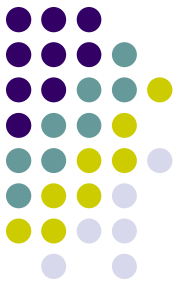
Internal: **sensory representation**
(of stimuli and experiences)

sensitivity
(for wasps)

preparations
(for actions)

Relationships between Concepts



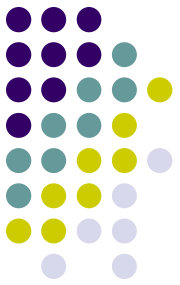


The Detailed Model

Formalisation of the concepts:

External behavioural level		
observation		
of stimuli	(wasp close)	
of experiences	(being stung)	
action	(panic)	
	(scream ouch!)	
Internal cognitive level		
sensory representation	(of wasp close)	
	(of being stung)	
sensitivity	(for wasp)	
preparation	(for panicking)	
	(for screaming ouch!)	

Internal Dynamic Relationships for Adaptive Behaviour (1)



IAB1

at any point in time

if the individual observes something,
(e.g., wasp close, being stung)

then after this the corresponding sensory
representation holds

observed(X) → sr(X)

IAB2

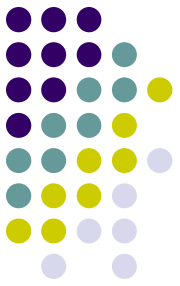
at any point in time

if the sensory representation for being stung holds,
then after this pain holds

sr(stung) → pain

VARIABLE

Internal Dynamic Relationships for Adaptive Behaviour (2)



IAB3

at any point in time

if pain holds

and the sensory representation of a wasp close holds,

then after this $s(1)$ holds

$\text{pain} \wedge \text{sr}(\text{wasp_close}) \longrightarrow \text{s}(1)$

IAB4

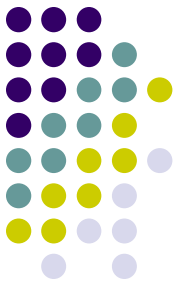
at any point in time

if $s(1)$ holds,

then at all later time points $s(1)$ holds

$s(1) \longrightarrow \text{s}(1)$

Internal Dynamic Relationships for Adaptive Behaviour (3)



IAB5

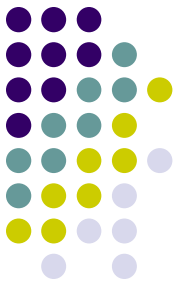
at any point in time

if $s(1)$ holds,

and the sensory representation of a wasp close holds,
then the individual will prepare for panic

$s(1) \wedge sr(\text{wasp_close}) \longrightarrow \text{prepfor(panic)}$

Internal Dynamic Relationships for Adaptive Behaviour (4)



IAB6

at any point in time

if the individual is prepared for an action
(e.g., panic, “ouch”),

then the individual will perform that action

prepfor(X) →→ performed(X)

IAB7

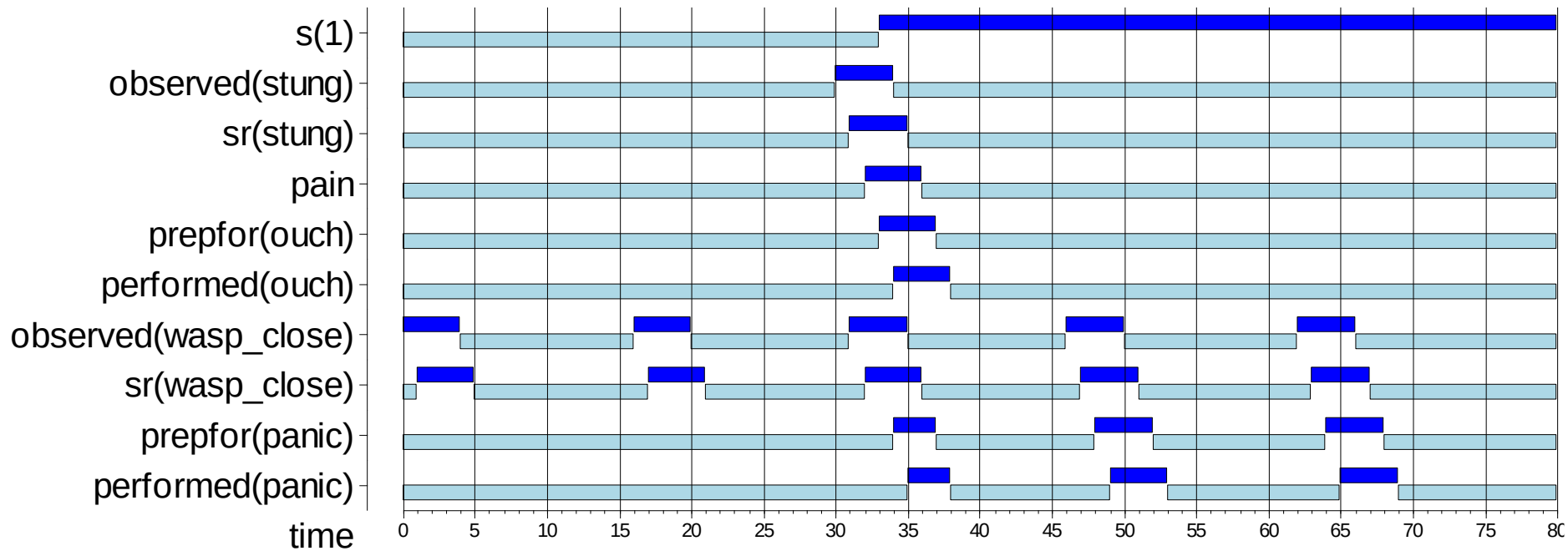
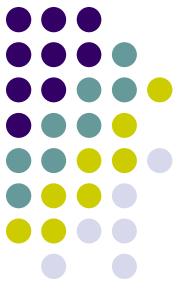
at any point in time

if pain holds,

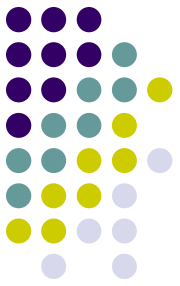
then the individual will prepare for screaming “ouch!”

pain →→ prepfor(ouch!)

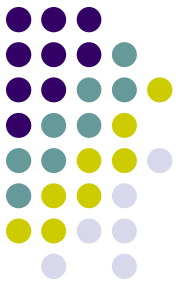
Example Simulation Trace



Gradual and Relative Adaptive Behaviour



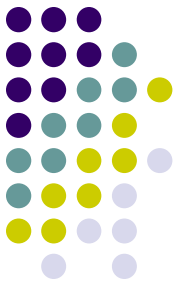
- Sensitivity increases due to **a number of sting** events
- The **more** often a wasp has stung the individual, the **more** panic the individual will show, when a wasp is close.



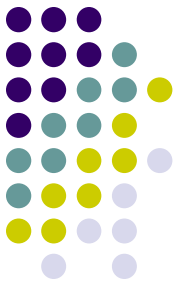
Chapter 12

MIND EXTENDED IN THE WORLD

Environmental Complexity Thesis

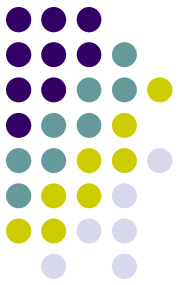


- **stimulus-response** behaviour
 - works only with visible food
- **delayed response** behaviour
 - food becoming invisible is taken into account
- **motivation-based** behaviour
 - food becoming absent is taken into account
 - never visible food is taken into account
- **adaptive behaviour**
 - accommodate behaviour to environment
- **extended mind behaviour**
 - environment participates in the behaviour



Extended Mind in World

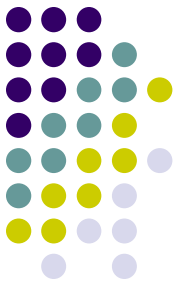
- Extended mind state for an individual:
A physical state in the external world used by the individual in its mental processes
- created by an action of the individual
- observed by the individual



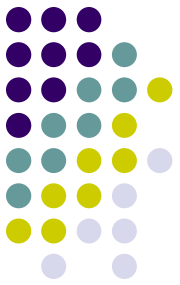
Extended Mind: Examples

- Tom Thumb (*Klein Duimpje*)
 - while walking, put pebbles on the ground regularly
 - to find the way back, follow the pebbles.
- Making lists
 - while having something to be done in mind, or something desired, write it down on the proper list
 - when free look at the lists to see what to do

Extended Mind: Characteristics

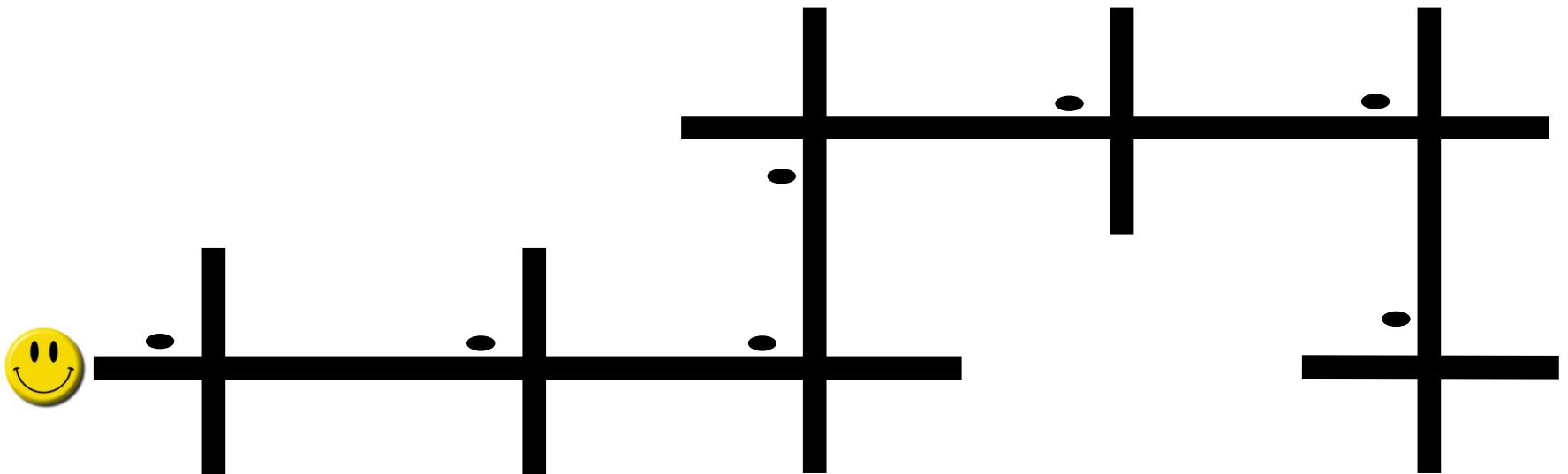


- Environment participates in individual's mental processes
- Individual's **mental processes** are **simplified**
- More intensive **interaction** with the world
- Individual **depends** on external world:
 - delegation of some mental representations
 - delegation of some mental capabilities
 - if Tom Thumb's pebbles are lost, he is lost.
 - if your to do list is lost, you are lost !

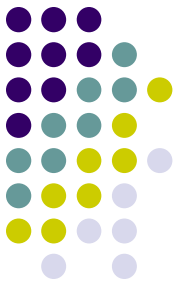


Case Study: Tom Thumb

- Drops a **pebble** before each crossing
- On way back, follows the pebbles



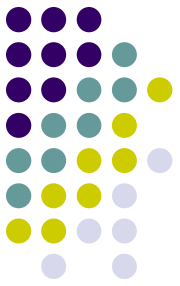
Specification of Input and Output



Input: **observation**
(of a crossing ahead)
(of a pebble at a certain position)

Output: **action**
(drop a pebble)
(go to a position)
(change direction)

Specification of External World States



External: **world fact**

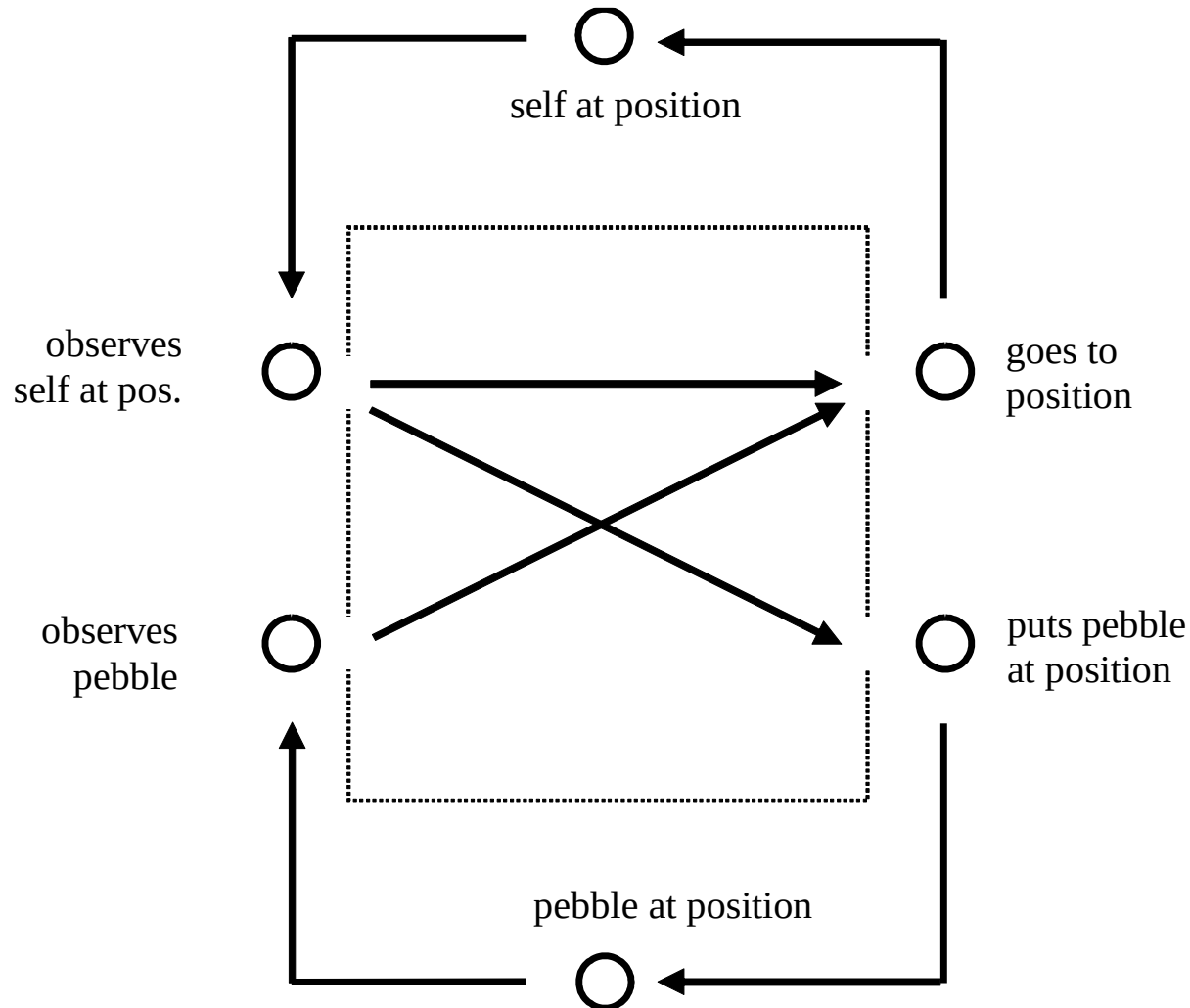
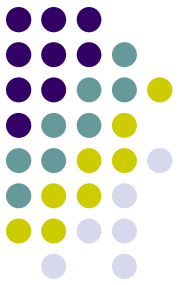
(of a pebble at a certain position)

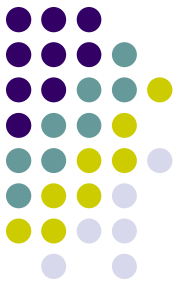
(of ourselves being somewhere)

(of which crossings are connected)

(of our direction)

Relationships between Concepts



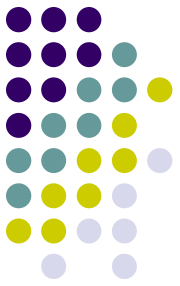


The Detailed Model

Formalisation of the concepts:

observation	(of position of self) (of position of pebble)	observed(at_position(self, P)) observed(at_position(pebble, P))
action performance	(drop pebble) (go to position) (change direction)	performed(drop_pebble) performed(goto(P)) performed(turn_direction)
pebble at a position in external world self at a position in external world adjacent positions in external world direction forward direction backward		world_fact(at_position(pebble, P)) world_fact(at_position(self, P)) world_fact(adjacent(P1, P2)) world_fact(direction_forward) world_fact(direction_backward)

Internal Dynamic Relationships for Extended Mind Behaviour (1)



LP1 (Dropping a pebble)

at any point in time

if the individual observes its position P

and the direction is forward

and there is no pebble at this position P

and there was no pebble dropping action yet

and P is not position p-END

then it will drop a pebble

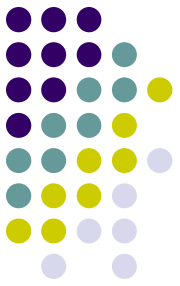
$\text{observed}(\text{at_position}(\text{self}, P) \wedge \text{world_fact}(\text{direction_forward}) \wedge \text{not}$
 $\text{world_fact}(\text{at_position}(\text{pebble}, P)) \wedge \text{not performed}(\text{drop_pebble}) \wedge$
 $P \neq \text{p-END}$



$\text{performed}(\text{drop_pebble})$

VARIABLE

Internal Dynamic Relationships for Extended Mind Behaviour (2)



LP2 (Moving forward)

at any point in time

if the individual observes its position P1

and it observes that P2 is an adjacent position

and the direction is forward

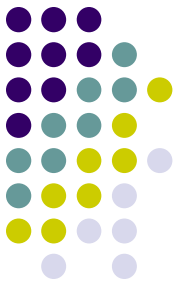
and it did not go to P2 yet

and it did not arrive at P2 yet

then the individual will go to P2

$$\text{observed}(\text{at_position}(\text{self}, P1)) \wedge \text{observed}(\text{adjacent}(P1, P2)) \wedge$$
$$\text{world_fact}(\text{direction_forward}) \wedge \text{not performed}(\text{goto}(P2)) \wedge \text{not}$$
$$\text{world_fact}(\text{at_position}(\text{self}, P2)) \longrightarrow \text{performed}(\text{goto}(P2))$$

Internal Dynamic Relationships for Extended Mind Behaviour (3)



LP3 (Turning around)

at any point in time

if the individual observes its position p-END

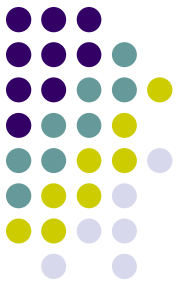
and the direction is not backward

and it did not just turn direction

then the individual will turn direction

$\text{observed}(\text{at_position}(\text{self}, \text{p-END})) \wedge \text{not}$
 $\text{world_fact}(\text{direction_backward}) \wedge \text{not performed}(\text{turn_direction})$
 $\longrightarrow \text{performed}(\text{turn_direction})$

Internal Dynamic Relationships for Extended Mind Behaviour (4)



LP4 (Moving backward to pebbles)

at any point in time

if the individual observes a pebble at a position P2

and the direction is backward

and it observes its own position P1

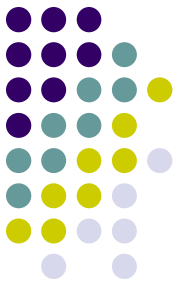
and it observes that P2 is an adjacent position

and it did not go to P2 yet

then the individual will go in the direction of that position

$$\text{observed}(\text{at_position}(\text{pebble}, P2)) \wedge \text{world_fact}(\text{direction_backward}) \wedge$$
$$\text{observed}(\text{adjacent}(P2, P1)) \wedge \text{observed}(\text{at_position}(\text{self}, P1)) \wedge \text{not}$$
$$\text{performed}(\text{goto}(P2)) \longrightarrow \text{performed}(\text{goto}(P2))$$

External Dynamic Relationships for Extended Mind Behaviour (1)



EP1 (Action effect for pebble dropping)

at any point in time

if the individual is at a certain position

and a pebble is dropped

then this pebble will be at this position

$\text{world_fact}(\text{at_position}(\text{self}, P)) \wedge \text{performed}(\text{drop_pebble}) \longrightarrow$
 $\text{world_fact}(\text{at_position}(\text{pebble}, P))$

EP2 (Persistence of pebbles)

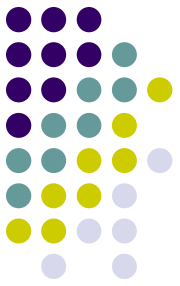
at any point in time

if a pebble is at a certain position

then at all later time points this pebble will be at this position

$\text{world_fact}(\text{at_position}(\text{pebble}, P)) \longrightarrow$
 $\text{world_fact}(\text{at_position}(\text{pebble}, P))$

External Dynamic Relationships for Extended Mind Behaviour (2)



EP3 (Action effect for going to a position)

at any point in time

if the individual goes to a certain position

then he will be at this position

`performed(goto(P)) → world_fact(at_position(self, P))`

EP4 (Action effect for turning around)

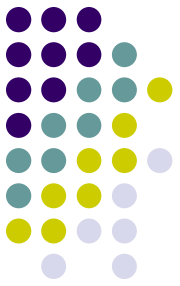
at any point in time

if the individual turns direction

then the direction will be backward

`performed(turn_direction) → world_fact(direction_backward)`

External Dynamic Relationships for Extended Mind Behaviour (3)



EP5 (Persistence of forward direction)

at any point in time

if the individual has forward direction

and he is not at position p-END

then he will have this direction

$\text{world_fact}(\text{direction_forward}) \wedge \text{not world_fact}(\text{at_position}(\text{self}, \text{p-END})) \longrightarrow \text{world_fact}(\text{direction_forward})$

EP6 (Persistence of backward direction)

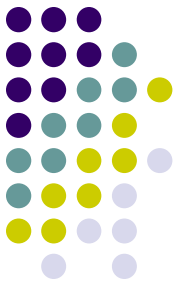
at any point in time

if the individual has backward direction

then he will have this direction

$\text{world_fact}(\text{direction_backward}) \longrightarrow \text{world_fact}(\text{direction_backward})$

External Dynamic Relationships for Extended Mind Behaviour (4)



EP7 (Persistence of being at a position)

at any point in time

if the individual is at a position P1

and for all positions Px he is not going to Px

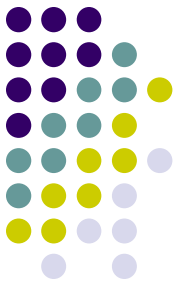
then he will be in position P1

`world_fact(at_position(self, P1)) ∧`
`[∀Px:Positions not performed(goto(Px))]`

→
`world_fact(at_position(self, P1))`

“for all” quantifier: the expression holds for ALL instantiations of Px:
`not performed(goto(p1)) ∧ not performed(goto(p2)) ∧ not performed(goto(p3))`

External Dynamic Relationships for Extended Mind Behaviour (5)



EP8 (Observation of pebble at own position)

at any point in time

if the individual is at a certain position
and a pebble is present at this position
then he will observe the pebble at this position

$\text{world_fact}(\text{at_position}(\text{self}, P)) \wedge \text{world_fact}(\text{at_position}(\text{pebble}, P)) \longrightarrow$
 $\text{observed}(\text{at_position}(\text{pebble}, P))$

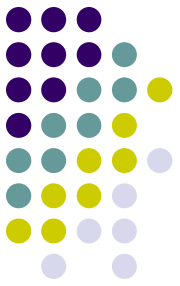
EP9 (Observation of pebble at adjacent position)

at any point in time

if the individual is at a certain position P1
and a pebble is present at adjacent position P2
and the direction is backward
then he will observe the pebble at this position

$\text{world_fact}(\text{at_position}(\text{self}, P1)) \wedge \text{world_fact}(\text{adjacent}(P2, P1)) \wedge$
 $\text{world_fact}(\text{at_position}(\text{pebble}, P2)) \wedge \text{world_fact}(\text{direction_backward}) \longrightarrow$
 $\text{observed}(\text{at_position}(\text{pebble}, P2))$

External Dynamic Relationships for Extended Mind Behaviour (6)



EP10 (Observation of self)

at any point in time

if the individual is at a certain position P

then he will observe that he is at this position

`world_fact(at_position(self, P)) →→ observed(at_position(self, P))`

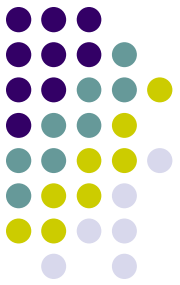
EP11 (Observation of adjacent positions)

at any point in time

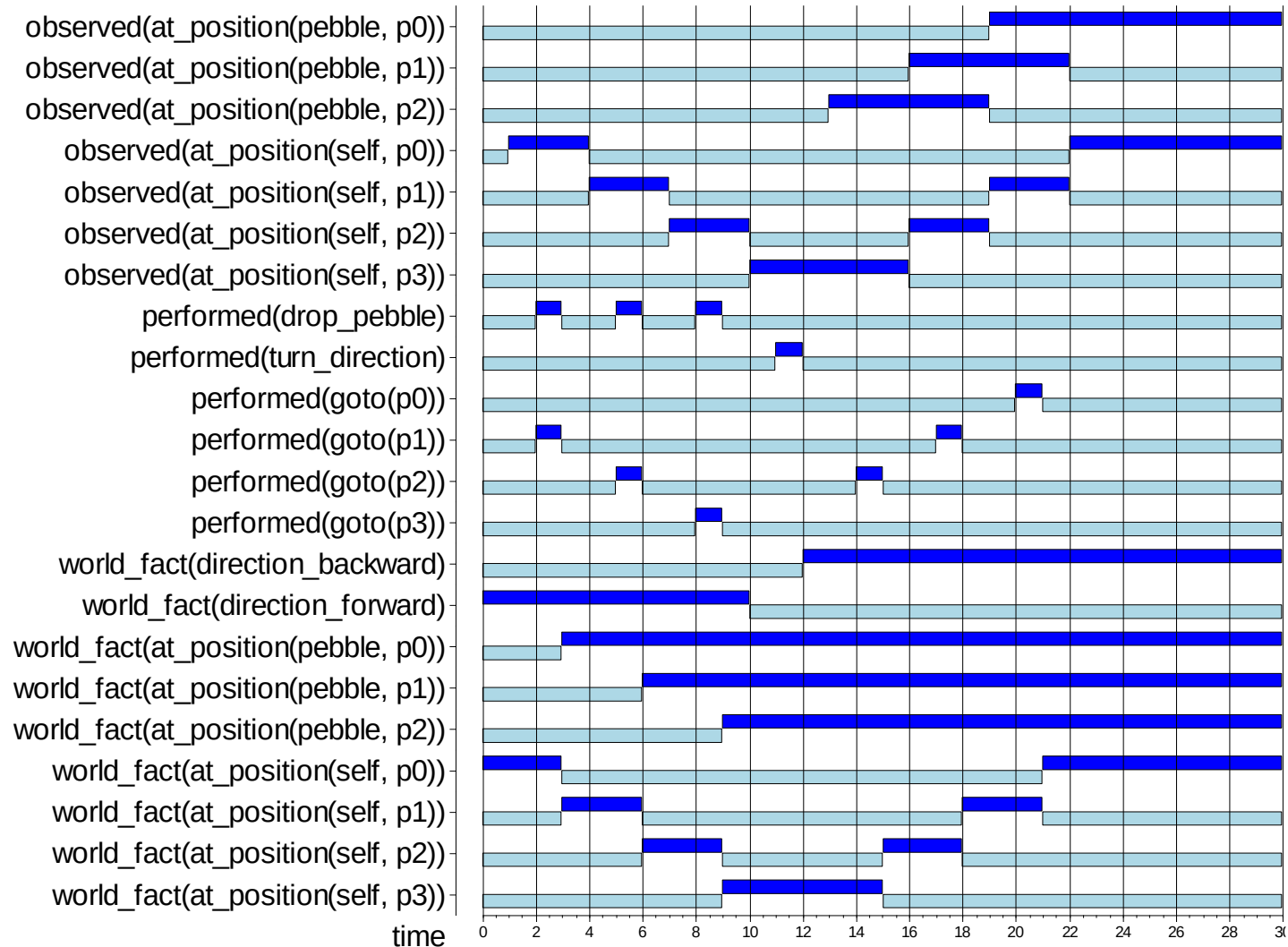
if positions P1 and P2 are adjacent

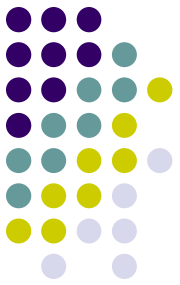
then the individual will observe this

`world_fact(adjacent(P1, P2)) →→ observed(adjacent(P1, P2))`



Example Simulation Trace

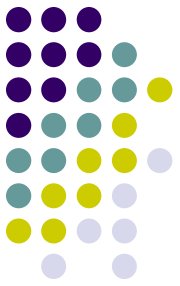




Evaluation

Assumptions made:

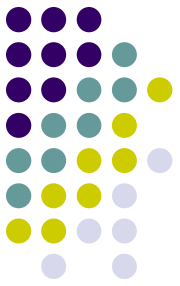
- Pebbles and their positions **persist**.
- **No other pebbles** are there than the ones put by Tom Thumb.
- Pebbles are **always found back** by observation.
- The way **back** is clearly **distinguished** from the way **forth**. For example, on the way forth no pebbles are followed, and on the way back no pebbles are dropped.



Further Explorations

- **Bread crumbs** instead of pebbles.
- Shared extended mind by **multiple individuals**.
- Behaviour based on “to-do lists”.
- Pheromone-based behaviour of ants.

Bread crumbs instead of pebbles



EP12 (Conditional persistence of bread crumbs)

at any point in time

if a piece of bread is at a certain position

and no animal is present at this position

then this piece of bread will be present at this position

$\text{world_fact}(\text{at_position}(\text{bread}, P)) \wedge \text{not}$
 $\text{world_fact}(\text{at_position}(\text{animal}, P))$



$\text{world_fact}(\text{at_position}(\text{bread}, P))$